Finite length of ordinary alternating regret matching in potential games

Abstract

Ordinary regret matching retains signed cumulative regrets, including negative entries that can delay a strategy change for a long time. We prove that its deterministic cyclic trajectory in every finite exact potential game has finite total strategy length and converges to one Nash equilibrium, from every initial mixed profile with zero cumulative regrets. The positive parts of the regret vectors also have finite total length. A one-step potential budget controls the extra contribution from coordinates emerging from negative memory. This yields a weighted gradient estimate for the polynomial potential lifted to a fixed product of spheres. The analytic gradient inequality then makes the square roots of the cycle gains summable. The result concerns the current profiles and does not assert bounded signed regrets or a uniform complexity bound; the known exponential transient remains compatible with convergence.

Keywords: regret matching; potential games; Nash equilibrium; finite length; analytic gradient inequality

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1 The cyclic recurrence and its limit

Let n players have finite action sets of sizes $m_i \geq 2$, and write

$$X = \prod_{i=1}^{n} \Delta_{m_i}, \qquad \Delta_m = \{x \in \mathbb{R}^m : x \geq 0,\ \langle \mathbf{1}, x\rangle = 1\}. \tag{1}$$

For a mixed profile $x \in X$, let $u_i(x_{-i}) \in \mathbb{R}^{m_i}$ be player i's full expected utility vector. Assume that the game has an exact potential. Its multilinear extension Φ satisfies

$$\Phi(z_i, x_{-i}) - \Phi(x_i, x_{-i}) = \langle z_i - x_i, u_i(x_{-i})\rangle \qquad (z_i \in \Delta_{m_i}). \tag{2}$$

All payoff and potential entries are finite real numbers. Put

$$\Delta_\Phi = \max_X \Phi - \min_X \Phi, \qquad m_* = \max_i m_i. \tag{3}$$

The symbol $[r]_+$ denotes the coordinatewise positive part of a real vector.

Initialize any $x^0 \in X$ and set $r_i^0 = 0$ for every player. In each cycle, visit the players in the fixed order $1, \ldots, n$. At player i's activation use the current profile, including the earlier updates in that cycle, and perform

$$g_i = u_i(x_{-i}) - \langle x_i, u_i(x_{-i})\rangle \mathbf{1}, \qquad r_i^+ = r_i + g_i, \tag{4}$$

$$\theta_i^+ = [r_i^+]_+, \qquad x_i^+ = \begin{cases} \theta_i^+ / \|\theta_i^+\|_1, & \theta_i^+ \neq 0, \\ x_i, & \theta_i^+ = 0. \end{cases} \tag{5}$$

All other blocks remain fixed at this activation. The signed state r_i in (4) is never truncated or reset. Retention in the second line of (5) is part of the algorithm. We write x^k, r^k, θ^k for cycle endpoints, with $\theta_i^k = [r_i^k]_+$.

This is ordinary alternating regret matching, with the strategy indexed after its corresponding regret update. The equivalent output-before-feedback convention appears in Algorithm 1 and Section 2.2 of Anagnostides et al. [1]. In particular, the statement below concerns deterministic full-information cyclic updates.

Theorem 1. *For every finite exact potential game, every initial mixed profile, and the zero signed initialization above, the potential is nondecreasing at every activation, and each $\|\theta_i\|_2$ is nondecreasing. At every cycle endpoint,*

$$\sum_{i=1}^{n} \frac{\|\theta_i^k\|_2}{\sqrt{m_i}} \le \Phi(x^k) - \Phi(x^0) \le \Delta_\Phi, \tag{6}$$

and

$$\sum_{k=0}^{\infty}\sum_{i=1}^{n} \|\theta_i^{k+1} - \theta_i^k\|_2^2 \le (\Delta_\Phi)^2 \sum_{i=1}^{n} m_i. \tag{7}$$

Moreover, both lengths are finite:

$$\sum_{k=0}^{\infty}\sum_{i=1}^{n} \|x_i^{k+1} - x_i^k\|_2 < \infty, \tag{8}$$

$$\sum_{k=0}^{\infty} \left(\sum_{i=1}^{n} \|\theta_i^{k+1} - \theta_i^k\|_2^2 \right)^{1/2} < \infty. \tag{9}$$

The current profiles x^k converge to one Nash equilibrium. The full sequence of intermediate activation profiles has the same limit and the finite strategy length in (8).

The theorem imposes no eventual support condition and no lower bound on individual positive probabilities. It asserts no boundedness or convergence of the full signed state, and no numerical convergence rate. Its finite-length constants can depend on the initialized trajectory.

1.1 The source question and method ancestry

The final ICLR 2026 paper [1, Section 5] asks whether ordinary regret matching converges asymptotically even with alternating updates. Theorem 1 answers that question for its cyclic recurrence and zero regret initialization, and strengthens convergence to finite length. The exponential transient in [1, Corollary 4.5] remains consistent with this qualitative conclusion.

Regret matching+ uses the different update $r_i^+ = [r_i + g_i]_+$. The potential and state-norm accounting developed for that recurrence in [1, 2] is useful ancestry. For the signed

recurrence, coordinates that become positive after accumulating negative memory create an additional nonnegative term. We retain this term and use it to control their weighted gradient contribution. The analytic ingredient is the classical gradient inequality in [3, Lemma 2.1]; the recurrence-specific step is the bound at actual cyclic endpoints proved below.

Earlier no-regret and fictitious-play results for potential games, including [4, Theorem 1 and Section 3.4], concern empirical belief variables. The present conclusion concerns the current mixed strategy profiles generated by (4)–(5). This comparison does not claim an exhaustive priority survey.

2 The potential budget with signed memory

Suppress the player index during one activation. Write

$$\begin{aligned} r' = r + g, \quad \theta = [r]_+, \quad \theta' = [r']_+, \quad d = \theta' - \theta, \\ R = \|\theta\|_2, \quad R' = \|\theta'\|_2, \quad s' = \|\theta'\|_1, \quad D = \langle x' - x, u \rangle. \end{aligned} \tag{10}$$

At every activation the current strategy is the normalization of θ whenever $\theta \neq 0$. Hence

$$\langle \theta, g \rangle = 0. \tag{11}$$

The equality is also valid when $\theta = 0$.

Lemma 1. *Let*

$$B = \{a : r_a \leq 0,\ r'_a > 0\}, \qquad Z = \sum_{a \in B} (-r_a)\theta'_a \geq 0. \tag{12}$$

Then

$$\langle \theta, d \rangle \geq 0, \qquad R'^2 - R^2 \geq \|d\|_2^2, \tag{13}$$

$$\langle \theta', g \rangle = \langle \theta', d \rangle + Z. \tag{14}$$

If $s' > 0$, then

$$s'D = \frac{R'^2 - R^2 + \|d\|_2^2}{2} + Z \geq \|d\|_2^2 + Z. \tag{15}$$

If $s' = 0$, then $\theta = \theta' = 0$ and $x' = x$. In all cases,

$$D \geq 0, \qquad R' - R \leq \sqrt{m}\, D. \tag{16}$$

Proof. On an old positive coordinate, $d_a = \max\{0, \theta_a + g_a\} - \theta_a \geq g_a$. Multiplication by θ_a and (11) give $\langle \theta, d \rangle \geq 0$, which proves (13) by expansion. On surviving positive coordinates, $g_a = d_a$. On newly positive coordinates, $d_a = \theta'_a = r_a + g_a$, so $g_a - d_a = -r_a$. Coordinates with $\theta'_a = 0$ contribute nothing to (14).

If $s' > 0$, then $\langle x, g\rangle = 0$ and $s'D = \langle \theta', g\rangle$; expansion gives (15). Also

$$s'D \geq \langle \theta', \theta' - \theta\rangle \geq R'(R' - R).$$

Since $s' \leq \sqrt{m}\, R'$, this proves (16). If $s' = 0$, (13) forces $R = 0$; the retention rule gives $D = 0$ without any division. ∎

At a player's activation, exactness (2) identifies D with the potential increment. Summing (16) over the actual unilateral updates proves (6), also at intermediate profiles. In particular,

$$\|\theta_i\|_2 \leq \sqrt{m_i}\Delta_\Phi, \qquad \|\theta_i\|_1 \leq m_i\Delta_\Phi. \tag{17}$$

Telescoping (13) for each player proves (7). This completes the budget part of Theorem 1.

If $\Delta_\Phi = 0$, every unilateral gain is zero by (2). The initial profile is Nash, all advantages vanish, and the trajectory is constant. Henceforth assume $\Delta_\Phi > 0$ and put

$$W = m_*\Delta_\Phi > 0. \tag{18}$$

Every post-update s' is at most W.

3 A weighted gradient estimate at each activation

A positive part cannot return to zero after becoming nonzero, by (13). During a player's initial phase with zero positive part, its strategy remains x_i^0. All regret increments have zero inner product with that fixed strategy. Thus zero initialization implies

$$\langle x_i^0, r_i\rangle = 0 \quad \text{while } \theta_i = 0. \tag{19}$$

The entries of r_i are then nonpositive, so $r_i[a] = 0$ on $\mathrm{supp}(x_i^0)$. At a zero-to-zero activation the same holds before and after the update, giving

$$g_i[a] = 0 \quad \text{on the support of the retained strategy.} \tag{20}$$

This does not say that all advantage coordinates are nonpositive. A positive advantage can still be hidden behind negative signed memory.

There is also a fixed normalization constant for the entire trajectory. For each player that ever has a nonzero positive part, take the norm of its first such positive part, and let δ be the minimum of these finitely many positive numbers and 1. If no player ever becomes positive, set $\delta = 1$. Then

$$\|x' - x\|_2 \leq \kappa\|d\|_2, \qquad \kappa = \frac{2\sqrt{m_*}}{\delta} < \infty \tag{21}$$

holds at every activation. Indeed, when both positive parts are nonzero,

$$x' - x = \frac{d}{s'} + \frac{s - s'}{s'}x, \qquad |s - s'| \leq \sqrt{m}\|d\|_2, \qquad s' \geq \delta.$$

For a zero-to-positive update, $\|d\|_2 \geq \delta$ and the simplex diameter is $\sqrt{2}$; for a zero-to-zero update both movements vanish. The constant δ is independent of time for this trajectory, but it need not have an effective uniform lower bound over games or initial profiles.

Exactness implies that $u_i(x_{-i})$ and $\nabla_i \Phi(x)$ differ by an action-independent scalar. Their centered advantages therefore agree. Fix finite constants $L, G > 0$ such that $\nabla\Phi$ is L-Lipschitz on X and each block's coordinate range is at most G. These constants exist because Φ is a polynomial and X is compact and convex. In particular, $|g_a| \leq G$ at each activation.

Lemma 2. *For the utility vector used at an actual activation and its post-update strategy,*

$$\sum_a x'_a \big(u_a - \langle x', u\rangle\big)^2 \leq (W + G)D. \tag{22}$$

Proof. Suppose first that $s' > 0$. On the support of θ' outside B in (12), $g_a = d_a$. Hence

$$\sum_{a \notin B} x'_a g_a^2 \leq \|d\|_2^2 \leq s'D \leq WD. \tag{23}$$

On B, $g_a = d_a - r_a \geq d_a = \theta'_a > 0$, and

$$\sum_{a \in B} \theta'_a g_a = \sum_{a \in B} d_a^2 + Z \leq s'D. \tag{24}$$

Since $g_a \leq G$, this gives

$$\sum_{a \in B} x'_a g_a^2 \leq \frac{G}{s'} \sum_{a \in B} \theta'_a g_a \leq GD.$$

Thus $\sum_a x'_a g_a^2 \leq (W+G)D$. Subtracting the weighted mean $\langle x', g\rangle = D$ can only decrease this second moment, yielding (22). If $s' = 0$, then $x' = x$, $D = 0$, and (20) makes the weighted second moment zero as well. ∎

The nonnegative charge Z controls newly active coordinates regardless of the size of their previous negative regrets. The weight x'_a allows individual positive probabilities to approach zero.

4 A fixed analytic potential and the cyclic timing

Consider the compact product of full unit spheres

$$\mathcal{S} = \prod_{i=1}^{n} S^{m_i - 1}, \qquad F(y) = \Phi(y_1^{\odot 2}, \ldots, y_n^{\odot 2}), \qquad y_i^k = \sqrt{x_i^k}. \tag{25}$$

Here squares and the nonnegative square root are coordinatewise. The restriction F is analytic on $\mathcal{S}$. The domain does not change when strategy supports change, and F is

smooth where coordinates vanish. We do not differentiate the inverse square-root map. The Riemannian gradient for the induced Euclidean metric is

$$\begin{aligned}\operatorname{grad}_i F(y) &= 2(I - y_i y_i^T)\operatorname{diag}(y_i)\nabla_i\Phi(x)\\ &= 2\operatorname{diag}(y_i)\big(\nabla_i\Phi(x) - \langle x_i, \nabla_i\Phi(x)\rangle \mathbf{1}\big),\end{aligned} \tag{26}$$

where $x_i = y_i^{\odot 2}$. The norm of the linear operator preceding $\nabla_i\Phi$ in the first line is at most 2.

Write $x^{k,0} = x^k$, and let $x^{k,i}$ be the profile after player i updates during cycle k, so $x^{k,n} = x^{k+1}$. Define

$$\begin{gathered}D_i^k = \Phi(x^{k,i}) - \Phi(x^{k,i-1}) \geq 0, \qquad \gamma_k = \Phi(x^{k+1}) - \Phi(x^k) = \sum_i D_i^k,\\ d_i^k = \theta_i^{k+1} - \theta_i^k.\end{gathered} \tag{27}$$

Since each block moves once per cycle, (15) and (21) imply

$$\sum_i \|d_i^k\|_2^2 \leq W\gamma_k, \qquad \|x^{k+1} - x^{k,i-1}\|_2^2 \leq \kappa^2 W\gamma_k. \tag{28}$$

For the second inequality, only blocks $j \geq i$ differ, and the squared product norm is the sum of their squared actual movements.

Lemma 3. *At the actual cycle endpoints,*

$$\|\operatorname{grad} F(y^{k+1})\|_2^2 \leq C_0^2\gamma_k, \qquad C_0^2 = 8\big(W + G + nL^2\kappa^2 W\big). \tag{29}$$

Proof. Player i's strategy at x^{k+1} is its post-activation strategy. Apply Lemma 2 to the gradient vector $\widehat{u}_i = \nabla_i\Phi(x^{k,i-1})$ used at that activation. Equations (22) and (26) bound the resulting lifted block gradient by

$$2\sqrt{W+G}\sqrt{D_i^k}.$$

Replacing $\widehat{u}_i$ by $\nabla_i\Phi(x^{k+1})$ changes it by at most

$$2L\|x^{k+1} - x^{k,i-1}\|_2 \leq 2L\kappa\sqrt{W}\sqrt{\gamma_k}.$$

Squaring with $(a+b)^2 \leq 2a^2 + 2b^2$ and summing over the players proves (29). ∎

The constant is finite along the entire initialized trajectory. It does not depend on its current support, its smallest active probability, or the magnitude of its negative signed regrets. The estimate uses the current-profile evaluation times of the cyclic algorithm.

5 Finite length and equilibrium identification

We first record explicitly how the analytic gradient inequality is used. The local theorem states that an analytic function f near a point z admits constants $c > 0$ and $\alpha \in [0, 1)$ with

$$|f(w) - f(z)|^\alpha \leq c\|\nabla f(w)\|_2$$

in a neighborhood; see [3, Lemma 2.1]. Only this analytic inequality is imported here.

Potential monotonicity gives $\Phi(x^k) \to \ell$ and $\gamma_k \to 0$. The cluster set Ω of y^k is nonempty and compact. On Ω, continuity and (29) imply $F = \ell$ and $\operatorname{grad} F = 0$. Near any $\bar{y} \in \Omega$, select in each block one coordinate of magnitude at least $1/\sqrt{m_i}$. On a smaller neighborhood its sign is fixed and its magnitude is at least $1/(2\sqrt{m_i})$. Solve for that coordinate as the signed square root of one minus the sum of the other squared coordinates. This gives an analytic sphere chart whose derivative norm is at most $2\sqrt{m_*}$.

Apply the local analytic inequality to F in that chart. The chart derivative is tangent to $\mathcal{S}$, so the chain rule bounds the coordinate gradient by $2\sqrt{m_*}\|\operatorname{grad} F\|$. A finite cover of Ω therefore yields constants $C_1 < \infty$ and $\alpha \in [1/2, 1)$ such that, on a neighborhood of Ω with $|F - \ell| < 1$,

$$|F(y) - \ell|^\alpha \le C_1 \|\operatorname{grad} F(y)\|_2. \tag{30}$$

Indeed, take the maximum of $1/2$ and the finitely many local exponents, and the maximum of their constants; for a number in $(0, 1)$ increasing its exponent decreases its power. Every sufficiently late y^k lies in this neighborhood, since otherwise compactness would supply a cluster point outside it. Thus (30) holds along the entire tail with one exponent and one constant, without support stabilization.

Lemma 4. *The square roots of the cycle gains are summable:*

$$\sum_{k=0}^{\infty} \sqrt{\gamma_k} < \infty. \tag{31}$$

Proof. Put $e_k = \ell - \Phi(x^k) \ge 0$ and $a_k = \sqrt{\gamma_k}$. If $e_K = 0$ at a finite index, then every subsequent gain is zero, and the assertion follows. Otherwise all $e_k > 0$. Combining (30) with the shifted estimate (29), for all sufficiently large k,

$$e_k^\alpha \le C_1 C_0 a_{k-1}. \tag{32}$$

In particular $a_{k-1} > 0$ on this tail. Define the increasing concave function

$$\psi(e) = \frac{C_1}{1 - \alpha} e^{1-\alpha}. \tag{33}$$

Since $e_k - e_{k+1} = a_k^2$, concavity and (32) imply

$$\psi(e_k) - \psi(e_{k+1}) \ge \psi'(e_k)(e_k - e_{k+1}) \ge \frac{a_k^2}{C_0 a_{k-1}}. \tag{34}$$

Consequently

$$2a_k \le a_{k-1} + C_0\big(\psi(e_k) - \psi(e_{k+1})\big). \tag{35}$$

Summing from a sufficiently large K through T gives

$$\sum_{k=K}^{T} a_k \le a_{K-1} - a_T + C_0\big(\psi(e_K) - \psi(e_{T+1})\big) \le a_{K-1} + C_0 \psi(e_K). \tag{36}$$

The finite prefix and the nonnegative tail prove (31). No division by an earlier zero gain has been used. ∎

Completion of Theorem 1. The budget and square-sum claims were proved in Section 2. Equations (28) and (31) yield

$$\sum_{k=0}^{\infty}\left(\sum_{i}\|d_i^k\|_2^2\right)^{1/2} \le \sqrt{W}\sum_{k=0}^{\infty}\sqrt{\gamma_k} < \infty. \tag{37}$$

By (21) and Cauchy–Schwarz over the players,

$$\sum_{k=0}^{\infty}\sum_{i}\|x_i^{k+1}-x_i^k\|_2 \le \kappa\sqrt{n}\sum_{k=0}^{\infty}\left(\sum_{i}\|d_i^k\|_2^2\right)^{1/2} < \infty. \tag{38}$$

Each block moves only once per cycle, so this is also the total length over all individual activations. Thus $x^k \to x^* \in X$, and all intermediate activation profiles tend to the same x^*.

It remains to identify the limit. Stationarity of the lifted potential alone would only enforce indifference on played supports. Suppose instead that an action a of player i has a strictly positive limiting advantage,

$$c = u_i(x_{-i}^*)[a] - \langle x_i^*, u_i(x_{-i}^*)\rangle > 0.$$

Continuity makes this action's actual advantage at least $c/2$ at every sufficiently late activation of player i. The signed recurrence (4) then forces $r_i[a] \to +\infty$, contradicting (17). Its signed value at any earlier finite time, however negative, is finite and cannot prevent this contradiction. Hence every limiting advantage is nonpositive. Their x_i^*-weighted mean is zero, so no unilateral deviation improves payoff. The limit is a Nash equilibrium. ∎

6 Scope of the qualitative conclusion

Zero signed initialization is used in (19), including for players whose positive state stays zero forever. No extension to arbitrary signed initial states is claimed here. The result also retains fixed cyclic updates and the prescribed zero-state rule; it makes no assertion about simultaneous or sampled-feedback trajectories.

The quantities controlled in (6) and (9) are the positive parts of the signed state. The distinction is necessary even at equilibrium: in a one-player two-action game with utilities $(1, 0)$, initial strategy $(1, 0)$, and $r^0 = 0$, the trajectory is fixed while $r^k = (0, -k)$. Thus bounded positive parts and finite strategy length coexist with an unbounded signed state.

Finally, the first positive norm δ, the entry into the analytic neighborhood, and its local exponent can depend on the game and trajectory. The proof supplies a finite-length conclusion for each permitted instance, rather than a uniform time bound for reaching

an approximate equilibrium. This is consistent with the exponential transient established in [1]. No finite length of the auxiliary square-root sequence is needed or asserted.

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